



Structure & Syllabus for Semester-V
BACHELOR OF COMPUTER APPLICATIONS (BCA)
Programme

Gujarat University
2025 - 2026

**As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR
UNDERGRADUATE PROGRAMMES, UGC**

&

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of

Education Department, Govt. of Gujarat

STRUCTURE FOR SEMESTER - 5
GUJARAT UNIVERSITY
Bachelor of Computer Applications (B.C.A.)
(as per NEP 2020)

COURSE:	Bachelor Of Computer Applications (B.C.A.)
MAJOR:	COMPUTER APPLICATIONS
MINOR:	1. COGNITIVE MATHS
	2. DATA SECURITY
	3. WEB TECHNOLOGIES

SEMESTER – 5

MAJOR

CODE	COURSE	CREDITS
DSC-C-BCA-351T	System Analysis: Design & Development (Internal Evaluation of SDP 1: Viva Voce)	4
DSC-C-BCA-352T	Problem Solving Using Python	4
DSC-C-BCA-353P	Practical Solutions Using Python	4

MINOR

CODE	COURSE	CREDITS
DSC-M-BCA-354T	Digital Marketing in E-Commerce	4
DSC-M-BCA-354P	Digital Marketing & Search Engine Optimization	4
	OR	
DSC-M-BCA-354T	Block Chain Technology	4
DSC-M-BCA-354P	Block Chain Technology Practical	4
	OR	
DSC-M-BCA-354T	Foundation of Mathematical Applications	4
DSC-M-BCA-355P	Web Applications Development using PHP	4

SKILL ENHANCEMENT COURSE

(Any ONE Course. Course can also be chosen from Basket)

SEC-BCA-356	Project Management using SCRUM	2
SEC-BCA-356	Web Skill Development tools (Git, GitHub, Google Docs, Google sheet)	2
SEC-BCA-356	Information Security	2
SEC-BCA-356	Concepts of IOT	2
SEC-BCA-356	Network Administration	2

TOTAL CREDITS 22

Semester – V

**BACHELOR OF COMPUTER
APPLICATIONS**

MINORS

- 1. COGNITIVE MATHEMATICS (CG)**
- 2. DATA SECURITY (DS)**
- 3. WEB TECHNOLOGIES (WTECH)**

Course Name: FUNDAMENTALS OF MATHEMATICAL APPLICATIONS

Course Code: DSC-M-BCA-354T

Credits: 4

Course Outcomes:

On the completion of the course students will:

- To understand basic principle of mathematics and discrete mathematics
- To improve logical thinking.
- To use principals and logics as supplement to develop other subjects.

UNIT	TOPIC/ SUBTOPIC	TEACHING HOURS	CREDITS
1	[1] SET THEORY: Introduction, Methods of representation, Examples of set , Various types of set [Finite set, infinite set, Empty set, Singleton set, Equivalence set , Equal set, Disjoin set, Sub set, Proper and improper sub set, power set] Set Operations [union, intersection , complement of set, Set difference , Symmetric difference] , Venn-diagrams , Properties of set operations[Commutative, associative, distributive, de morgan's law Cardinality , Cartesian product. [2] FUNCTION : Introduction & Definition , Domain , Codomain and range of function, Types of function:(1) on base of mapping: one-one , many one, constant, onto, into (2) on base of degree : constant , linear , quadratic , exponential , cubic , polynomial, (3) Graphs of functions.	15	01
2	[1] RELATIONS : Introduction , Universal and Empty relation, inverse of relations ,Total numbers of relations , Types of relations : Reflexive , Symmetric, Transitive , Antisymmetric , Asymmetric , Irreflexive Equivalence relations , Partition of set , Partial order relations , Hasse Diagrams of posets , Upper bound & least upper bound , Lower bound & Greatest lower bound . [2] Lattice & Boolean Algebra : Bounded lattice , complete lattice , Distributive lattice , Complemented lattice , Boolean Algebra , Properties of Boolean Algebra , Minterms and Maxterms , Sum of product & product of sum , Complete sum of product form.	15	01
3	[1] MATRIX Introduction, Definition , Notations , Types of Matrix [Row , column , rectangular , square, diagonal , scalar , identity , zero , upper triangular , lower triangular , singular, symmetric , skew symmetric , Transpose matrix] Arithmetic Operations : Addition , Subtractions , Multiplications , Rank of matrix , Inverse matrix , Solve the equations by using inverse matrix . [2] LOGIC AND PROPOSITIONAL CALCULUS Introduction , Propositions and compound propositions , Basic logical operations[Conjunction , Disjunctions, Negation] , Truth table , Tautologies and Contradictions , Law of the algebra of propositions [Idempotent , Associative , Commutative , Distributive , Identity , Compliment , Involution , De Mogen's]	15	01
4	[1]Permutation & Combination	15	01

	Introduction, Factorial notation, permutations of n different things [exclude circular permutations], permutations of things if some of them are identical, Restricted permutations. Combinations of r items out of n, Restricted combinations [2] Probability : Introduction, Basic terms, Probability of various events by using Addition and Multiplication rule of probability, Probability of an event if selection is with replacement or without replacement, Baye's theorem.		
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P.S. Use of Calculator by student is allowed

Reference Books :

1. Business Mathematics by D.C. SANCHETI & V.K. KAPOOR Sultan chand & Sons
2. Discrete Mathematics by SEYMOUR LIPSCHUTZ, MARC LARS LIPSON, VARSHA PATIL Mc Grew Hill
3. Business Mathematics BY P.MARIAPPAN Pearson
4. Fundamentals of Statistics by S. C. GUPTA Himalaya Publishing House
5. Discrete Mathematics and its Applications by Kenneth H. Rosen, Kamala Krithivasan..... Mc Grew Hill

Course Name: Web Application Development using PHP

Course Code: DSC-M-BCA-355 P

Credits: 4

Course Outcomes:

The aim of this course is to enable students to

- Get familiar with server-side language for web development.
- Learn basic concepts of PHP, functions for database operations using MySQL.
- Learn about AJAX in PHP for faster processing of web applications.
- Get familiar with validations.

Prerequisites:

- Basic knowledge of data types, HTML tags.

Unit-1 Introduction to PHP

Introduction: Basic PHP syntax, PHP comments, data types in PHP, PHP variables, Echo and print statement, PHP strings(functions), PHP math functions, PHP operators, PHP if...else statements, switch case, Loop (While, do while , for, for each), PHP functions , PHP arrays(Indexed, Associative, multidimensional), PHP form handling

Unit-2 Advanced PHP

MYSQL database, MySQL connect, Create database, create table, Insert data, Select data, Update data, Delete data, Display data, AJAX introduction, PHP and AJAX, Methods and properties of XMLHttpRequest object, Concept of Cookie and Session in PHP

Unit	Particulars	Hours	Credits
1	Basic Programs	30	1
	1. Write a PHP program to check whether the number entered is Positive / Negative or Zero entered by user		
	2. Write a PHP program to find maximum number from the two numbers entered by user		
	3. Write a PHP program that prints Fibonacci series till the number entered by user.		
	4. Accept a string from user. Write PHP program to check whether that entered string is palindrome or not.		
	5. Write a PHP program to accept a number from user and find its square and cube.		
	6. Write a PHP program to find sum of n numbers. Take input from user.		
	7. Accept a two numbers from user. Write a PHP program to Add, Subtract and Multiply the numbers and displays result.		
	8. Accept a number from user as "Radius" of a circle. Write PHP program find Area of a Circle. (Area = PI * radius * radius).		
	9. Write a PHP program to print all even numbers till the entered number.		

	10.	Create a function that accepts 3 numbers as parameters and check whether the sum of 3 numbers is prime or not. Note: Input should be collected from user using an HTML page.		
	11.	Create a function that accepts a number as argument and displays sum of digit of a number. E.g. If Number is 123 then sum of Digits is 6 i.e., 1+2+3 =6 Note: Input should be collected from user using an HTML page.		
	12.	Write a PHP program that changes color of the web page using switch case.		
2.	Session, Cookies, Arrays, Functions		30	1
	1.	Write a PHP script to set a cookie named "username" with the value "Welcome" with an expiration time of one hour and display its value.		
	2.	Write a PHP script to delete a cookie named "welcome".		
	3.	Write a PHP script to retrieve and display the value of the session variable "loginid".		
	4.	Write a PHP script to check if a cookie named "visited" exists. If it does, display a welcome message; otherwise, display a default message.		
	5.	Create an array with 5 elements and print all array elements.		
	6.	Create an array with 5 elements. Find and display minimum and maximum value from the array.		
	7.	Write a PHP program that deletes duplicate values from array.		
	8.	Write a PHP program that checks whether the element is exists in the array or not. Give acknowledgement from the same.		
	9.	Create an associative array and display its elements.		
	10.	Create an array named \$sub, assign five elements to it and display the elements assigned using for loop and foreach statement		
	11.	Create an array named \$student that stores 5 elements bounded to different keys and access the same using the key element.		
	12.	Write a program in PHP to demonstrate the use of multidimensional arrays.		
	13.	Write a program in PHP to sort the array of given 5 numbers in ascending and descending order.		
	14.	Write a program to count the total number of times a specific value appears in an array.		
	15.	Create two functions in PHP, parameterized and non parameterized for implementing string concatenation operation.		
3.	MySQL		30	1
	1.	Write a PHP script for that creates a database named "DB-1" in MySQL. And also create a table "myTable" in above created database "DB-1".		
	2.	Write a PHP script for creating a "Product" table in "MyDB" database with fields Pro_id, Pro_name, Pro_price, Qty,		

		Rate . Also display an acknowledgement for the same as "Table created successfully".		
3.		Create a form containing four input fields(Pro_id, Pro_name, Pro_price, QOH) and Submit button. When the user clicks on the submit button, PHP script should be executed which inserts the record in the product table.		
4.		Create a form containing one input field (Product_id) and a Delete button. When the user clicks on the Delete button a PHP script should get executed and should delete the record of the product for the Product_id specified.		
5.		Create a form containing two input fields (Product_id, Rate) and Update button. When the user clicks on the Update button, the Rate of the Product_id specified should get updated by 5% of the current rate using a PHP script.		
6.		Create a form containing one input field (Product_id) and a search button. When the user clicks on the Search button a PHP script should get executed and should display the details of the product for the Product_id specified.		
7.		Write a PHP script for creating a "Product" table in "MyDB" database with fields Pro_id, Pro_name, Pro_price, Qty, Rate . Also write code to update product price to 100 whose product id is <4 in Product table".		
8.		Write a PHP script for creating a "Student" table in "MyDB" database with fields Student_id, Student_name, Course, Age, MobileNo . Also display an acknowledgement for the same as "Table created successfully".		
9.		Write a PHP Script to update record (Stud_Address) of student table where sid=13		
10.		Write a PHP Script to Delete Record of Student Table by clicking on Delete Button by specifying Student ID.		
11.		Write a PHP Program to update a record with your own value into the table Student against each column (Student ID, Student Name, Course ID, and Mobile No). Note: (a) Input should be collected from user using an HTML page. (b) Make Sure Student ID, Course ID, and Mobile No must not be Negative. (c) Student Name should not be Blank.		
12.		Write a PHP Script to Update Record of Employee table (Empname, Empsalary, Empaddress) from Employee Table. Update Employee Name to "ABC" whose Employee number is 3.		
13.		Write a PHP script that creates a database named "Product_Details" in MySQL. Create an HTML form containing four input fields (Pro_id, Pro_name, Pro_price, and QOH) And Submit button. When the user clicks on the submit button a PHP Script should be Executed which inserts the record in the product table. Note: input should be collected from user using an html page and there should not be duplicate product id in the table.		

	14.	Write a PHP Program to update a record with your own value into the table Employee against each column (Employee ID, Employee Name, Job Code, Salary and Mobile No). Note: (a) Input should be collected from user using an HTML page. (b) Make Sure Employee ID, Salary, and Mobile No must not be Negative. (c) Employee Name should not be Blank. (d) Salary length must not be greater than 5 digits.		
4.	AJAX and Validation		30	1
	1.	Create a form containing one input field (MyName). When the user enters his/her name and press any key, the form should display a welcome message for the user. Implement using AJAX.		
	2.	Create a form to enter a string. Write a PHP program for converting a string into uppercase using AJAX.		
	3.	Create a form containing a combobox with some product names as items. Whenever a user selects a particular product from the combobox, that selected item should be printed on the page. (i.e. without pressing submit button). Implement using AJAX.		
	4.	Create a form containing a combobox with some color names as items. Whenever a user selects a particular color from the combobox, that selected item should be printed on the page. (i.e. without pressing submit button). Implement using AJAX.		
	5.	Create a form containing a radiobutton with some product names. Whenever a user selects a particular product from the radiobutton, Display on the page as "You have selected _____ product". Implement using AJAX.		
	6.	Create a form containing one input field (Number). When the user enters the number and as the key is released, the form should display "whether the number is odd or even" with a message to the user. Implement using AJAX.		
	7.	Write a program to validate a blank field and also validate the length of a string entered (i.e. minimum length of 5).		
	8.	Write a program to validate and Email ID using regular expression and by using DOM		
	9.	Write a program to validate a username field and also validate the length of the username entered (i.e. minimum length of 7 and maximum length of 12) using AJAX.		
	10.	Write a PHP program to print user data using AJAX. Make one drop down select user id and print specific user detail by matching user id.		
	11.	Create a form containing one input field (Number). When the user enters the number and as the key is released, the form should display " whether the number is greater than 50 or not " with a message to the user. Implement using AJAX		
	12.	Write a program that checks a particular student Id already exists in the Student Table (Stud_id, Stu_Name, MobileNo, Country) or not. If the entered student ID exists		

		then display a message that : "Student ID already exists....". (Using AJAX)		
	13.	Write a program that checks a particular mobile number already exists in the student (stuld, stu_name, mob, country) table or not. If mobile number exists then display a message "Mobile Number Already registered. Try another Mobile Number" . If it does not exist then add the data in the student table. Implement using AJAX.		

Textbook:

Developing Web Applications in PHP and AJAX

Publisher: McGraw Hill

By B M Harwani

Reference Books:

1. Teach yourself PHP, MySQL and Apache

Publisher: Pearson

By Julie C. Meloni

2. PHP Ajax Cookbook

Publisher: PACKT Publishing

Other Web References:

<https://www.w3schools.com/php/>

<http://www.learn-php.org/>

<https://www.tutorialspoint.com/php/>

Required Softwares:

1. Notepad++

2. XAMPP or WAMP server

Accomplishments of the student after completing the Course:

After completion of this course Student would be able to

- To setup and configure XAMPP
- To develop server side programs using PHP in dynamic web development.
- To debug a PHP/AJAX web application.
- To use AJAX for efficient and fast communication between client and server.
